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Edge Devices Friendly Self-Supervised Monocular Depth Estimation Via Knowledge Distillation

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***Abstract*—Self-supervised monocular depth estimation (MDE) has great potential for deployment in a wide range of applications, including virtual reality, autonomous driving, and robotics. Nevertheless, most previous studies focused on complex architectures to pursue better performance in MDE. In this letter, we aim to develop a lightweight yet highly effective self-supervised MDE model that can deliver competitive performance in edge devices. We introduce a novel MobileViT-based depth (MViTDepth) network that can effectively capture both local features and global information by leveraging the strengths of convolutional neural networks (CNNs) and a vision transformer (ViT). To further compress the proposed MViTDepth model, we employ knowledge distillation, which leads to improved depth estimation performance. Specifically, the self-supervised MDE MonoViT is used as a teacher model to construct the knowledge distillation loss for optimizing a student model. Experimental results on benchmark datasets demonstrate that the proposed MViTDepth significantly outperforms Monodepth2 in terms of parameters and accuracy, thereby indicating its superiority in application to edge devices.**

***Index Terms*—Deep learning for visual perception, autonomous vehicle navigation, lightweight monocular depth estimation.**

I. Introduction

MONOCULAR depth estimation (MDE) is a vital task in computer vision that aims at estimating the depth of a scene from a single image. Since accurate depth information is crucial for various real-world applications such as virtual reality [1], autonomous driving [2], and robotics [3], MDE has attracted substantial research interest over the past decade. There are two main categories of deep learning-based MDE methods: supervised and self-supervised depth estimations.

Supervised MDE methods [4], [5], [6] have achieved excellent performance by making continuous improvements. However, acquiring a large-scale dataset of scene images with per-pixel depth annotations for supervised learning is still an extremely challenging task. To overcome this problem, a self-supervised MDE method was first proposed in [7] to estimate the dense depth map of a scene from a single image without the need for ground truth depth information. The self-supervised MDE relies on synchronized stereo image pairs or monocular videos to obtain supervisory signals during training. Since the camera motion is known, methods based on stereo image pairs [7], [8], [9] only require a single depth network for training. However, methods based on monocular videos [10], [11], [12] require an additional pose network in training. Nevertheless, the latter has fewer restrictions due to the fact that acquiring unlabeled monocular videos is easier than acquiring stereo image pairs.

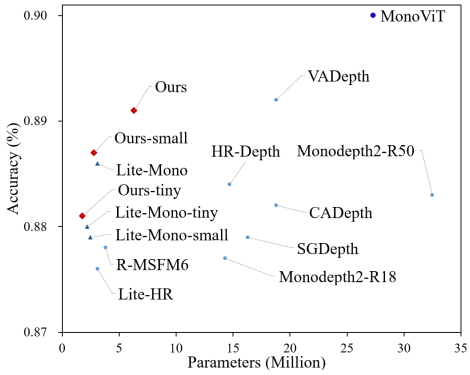


Fig. 1. The parameters and accuracy (δ_1) of different self-supervised methods for monocular depth estimation. All models are trained and tested on the KITTI dataset with the resolution of 192×640. Our proposed method achieves better trade-off between accuracy and model complexity.

Self-supervised methods have been proposed to enhance performance from diverse perspectives, like semantic segmentation [13], data pre-processing [14], and geometric priors [15]. These approaches largely benefit from an additional task or complex data processing. The network architecture also has a significant impact on the achievement of good performance in MDE. Most previous works [10], [12], [16] used the residual networks [17] as the backbone network. However, it is difficult for the convolution operation to perceive global information in such an architecture. Consequently, the vision transformer (ViT) [18] was introduced into the self-supervised MDE paradigm to capture long-range global information [11] [19]. However, ViT-based architectures that incorporate multi-head self-attention have significantly higher computational costs than ResNet-based models.

As mentioned above, we found that most existing works were devoted to pursuing better performance except of Lite-Mono [20]. With the increasing availability of edge devices, self-supervised MDE approaches can be widely deployed in numerous practical applications. Therefore, we aim at attaining the best balance between accuracy and model complexity for self-supervised MDE methods used in edge devices of

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demanding real-time depth estimation. By leveraging the strengths of convolutional neural networks (CNNs) and ViT, MobileViT [21] can effectively capture both local features and global information while simultaneously having low complexity and performing well in computer vision tasks. Accordingly, we introduce a novel MobileViT-based depth (MViTDepth) network to generate depth maps.

The essence of knowledge distillation is to transfer knowledge from a larger and more complex model (called a teacher model) to a smaller and simpler model (called a student model) with comparable performance. This technique is particularly useful in scenarios where computational resources are limited, such as mobile devices and edge computing. The scheme of offline-distillation is straightforward, and it efficiently transfers knowledge from a pre-trained teacher model to a student model. Since the self-supervised MDE MonoViT [11] achieved state-of-the-art (SOTA) performance, we employ its pre-trained model as the teacher network for offline-distillation. In this way, we can further simplify the complexity of our proposed MViTDepth network while maintaining satisfying performance. As depicted in Fig. 1, the proposed self-supervised MViTDepth achieves a much better trade-off between accuracy and model complexity than existing self-supervised MDE SOTA methods.

To summarize, our main contributions are as follows:

- A novel lightweight MViTDepth architecture is proposed for self-supervised MDE in terms of the balance between model complexity and performance.
- Knowledge distillation from a pre-trained MonoViT network is proposed, utilizing a logarithmic form of distillation loss to further unleash the potential of MViTDepth.
- Extensive experiments on benchmark datasets are carried out to confirm the superiority of our proposed lightweight model (of different sizes) for use in edge devices.

II. Related Work

A. Monocular Depth Estimation

Estimating depth only from a single image is an inherent ill-posed problem since the same 2D image can correspond to multiple different 3D scenes. Deep learning-based methods were proposed to tackle this issue, and they have shown promising results. Early attempts on deep learning-based MDE methods [4], [5] predominantly focused on supervised learning, and they promptly achieved SOTA performance. However, the performance of fully supervised methods heavily depends on the availability of large-scale ground truths, which are expensive to acquire and cannot involve various scenarios. Inspired by the view synthesis task [22], self-supervised methods that employ the image reconstruction loss as a supervisory signal were developed as an attractive alternative. Despite the stereo image pairs can be utilized to construct the image reconstruction loss [7], acquiring monocular videos is significantly less challenging than obtaining stereo image pairs. SfMLearner [20], [23], [24] adopted monocular videos to train self-supervised MDE models. Unlike the methods that utilize the stereo image pairs [7], [8], those based on monocular videos [15], [23] need to estimate camera ego-motion and depth simultaneously during training. Recently, various works [11], [13], [16], [19], [25], [26] investigated different techniques for increasing the accuracy of self-supervised MDE, including domain separation, uncertainty, and semantic segmentation guidance. Compared with introducing an additional task, it is more straightforward to improve the loss function in order to achieve competitive results. Monodepth2 [10] was developed to reduce the number of depth artifacts by using a multi-scale appearance matching loss, eliminate camera-related stationary pixels by adopting auto-masking, and address the occlusion problem by minimizing re-projection loss. Our work also follows the general pipeline of Monodepth2. The backbone network of Monodepth2 is based on ResNet [17], which is actually a CNN-based network with limited capability of extracting significant features from noise and perceiving long-range dependencies.

B. Vision Transformer

The ViT [18] has exhibited great potential in several computer vision tasks [27], [28]. In these tasks, an image is divided into many non-overlapping patches, and then a multi-head self-attention mechanism is used to learn the inter-patch representations in transformers [29]. Recently, ViT-based models have been widely used in MDE due to their excellent ability to achieve SOTA results [6], [11]. However, ViT-based architectures with large-scale parameters generally result in high computational overhead. In addition, existing lightweight methods for MDE [12], [20], [30] tend to have a high computational burden or low accuracy when evaluated on standard metrics. To make a compromise between model performance and complexity, MobileViT [21] was proposed to combine the strengths of a CNN and ViT. As a consequence, the MobileViT-based model is utilized as our backbone network in self-supervised MDE and allows us to meet lightweight and low computational requirements.

C. Knowledge Distillation

Knowledge distillation was originally proposed in [31] to transfer knowledge from a larger teacher model to a smaller student model by training on an identical task. Through this, we can enhance the performance of the student model by learning from the higher-level representations of the teacher model. This technique is often used to compress large and complex model into a small but effective model for resource-constrained applications. Moreover, knowledge distillation mainly has three categories [32]: offline-distillation, online-distillation, and self-distillation. Specifically, the training process of offline-distillation typically involves two stages. First, a teacher model is pre-trained on a large dataset using a standard supervised learning approach. Second, a student model is trained to learn from the output of the teacher model instead of using the ground truth labels of the dataset. This leads to a reduction in model size and improved computational efficiency while still preserving similar performance across models. Offline-distillation can thus be used to boost the capability of lightweight models. Recently, two self-supervised MDE methods based on knowledge distillation were proposed in [9] and [12], and they utilized self-distillation and offline-distillation, respectively. As shown in Fig. 1, MonoViT [11],

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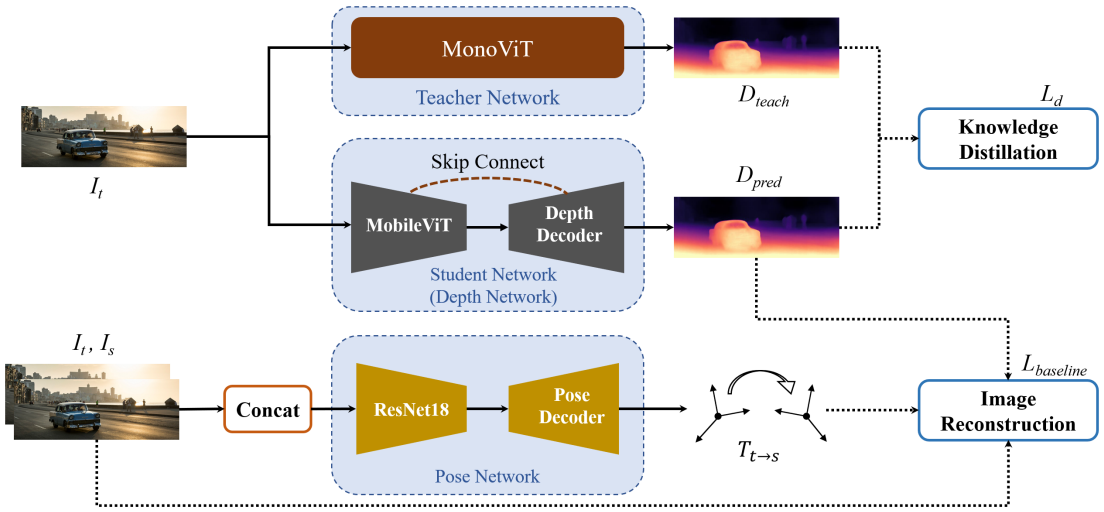


Fig. 2. The overview of MViTDepth pipeline. During training, we input the target frame I_t into the student network to predict a depth map D_{pred}, while input the target frame I_t into the teacher network that is pre-trained on KITTI to generate a depth map D_{teach}. Then, we concatenate the target frame I_t and source frame I_s as the input of pose network to produce relative pose $T_{t\to s}$ between I_t and I_s. Finally, we combine the knowledge distillation loss and reconstruction loss to optimize the depth network and pose network simultaneously. At test stage, we only employ the depth network to derive depth map.

as the current SOTA self-supervised MDE method, achieved top performance on various benchmark datasets. For the sake of simplicity and efficiency, offline-distillation is employed in our model. We first train the MonoViT model on the KITTI dataset, and then use it as the teacher model for our MViTDepth model.

III. Method

In this section, we first review the basic framework of self-supervised monocular depth estimation. Then, we present the knowledge distillation that can optimize the MViTDepth network further by constructing a distillation loss. Finally, we provide the details of the proposed MViTDepth network.

A. Self-Supervised Monocular Depth Estimation

Following previous work [10], given the continuously adjacent RGB frames I_t and I_s, $s \in \{t-1, t+1\}$, we can use the pose network based on ResNet [17] to obtain the relative pose $T_{t\to s}$ by concatenating the target frame I_t and source frame I_s, which is defined by

$$T_{t\to s} = PoseNet(I_t, I_s). \tag{1}$$

Meanwhile, the target frame I_t is also input into the depth network to predict a dense depth map:

$$D_{pred} = DepthNet(I_t). \tag{2}$$

When we obtain the target image depth map D_{pred} and the relative pose $T_{t\to s}$ between the target frame I_t and source frame I_s, then the target frame I_t pixel coordinates can be projected to its corresponding view in the source frame I_s, ultimately generating the warped image:

$$I_{s\to t} = I_s\langle KT_{t\to s}D_{pred}K^{-1}p_t\rangle, \tag{3}$$

where K represents the camera intrinsic matrix that can be computed a prior. Moreover, we assume that matrices K in all views are identical. Here, p_t is the homogeneous pixel coordinates of I_t, and $\langle\cdot\rangle$ is the sampling operator. We adopt the photometric re-projection error $pe(\cdot)$ to calculate the reconstruction loss:

$$L_{ph} = pe(I_t, I_{s\to t}). \tag{4}$$

Note that L_{ph} is consisting of two parts [10]: the structural similarity index measure (SSIM) and L_1 loss. Correspondingly, (4) can be reformulated as follows:

$$pe(I_t, I_{s\to t}) = \frac{\alpha}{2}\left[1-\text{SSIM}(I_t, I_{s\to t})\right] + (1-\alpha)\|I_t - I_{s\to t}\|_1, \tag{5}$$

where α represents a hyperparameter, and it is set to 0.85 in our experiments. Following [10], the auto-masking is also used to filter out the pixels that are stationary relative to the camera, which can be formulated as:

$$\mu = \left[\min_s pe(I_t, I_{s\to t}) < \min_s pe(I_t, I_s)\right], \tag{6}$$

where $[\cdot]$ denotes the Iverson bracket. As in [8], to ensure the smoothness of the estimated depth maps, we also employ the edge-aware smoothness loss, which can be defined as follows:

$$L_{sm} = |\partial_u d_t^*| \exp\{-|\partial_u d_t^*|\} + |\partial_v d_t^*| \exp\{-|\partial_v d_t^*|\}, \tag{7}$$

where $d_t^* = d_t\sqrt{d_t}$ represents the mean-normalized inverse depth [33] to prevent shrinking of the estimated depth maps. Therefore, the baseline loss can be expressed as:

$$L_{baseline} = \frac{1}{S}\sum_{i=1}^{S}\left(\mu L_{ph}^i + \lambda L_{sm}^i\right), \tag{8}$$

where S refers to the number of different scale outputs generated by the depth decoder, and λ is set to $1e^{-3}$ empirically.

B. Depth Distillation

As illustrated in Fig. 1, MonoViT achieved SOTA performance in self-supervised MDE. To enhance the accuracy of our model, we apply the offline-distillation scheme where knowledge is transferred from a pre-trained MonoViT model to our student depth network. Both MonoViT and our depth

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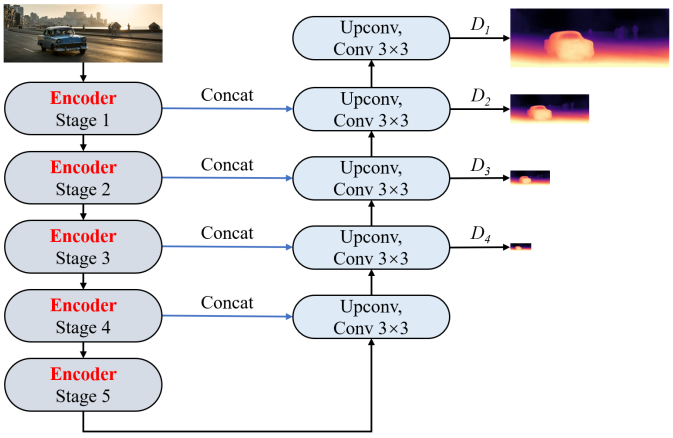


Fig. 3. Structure of student network. MobileViT is utilized as the depth encoder to extract features. The decoder concatenates the corresponding scale features produced by the encoder to generate the multi-scale depth maps.

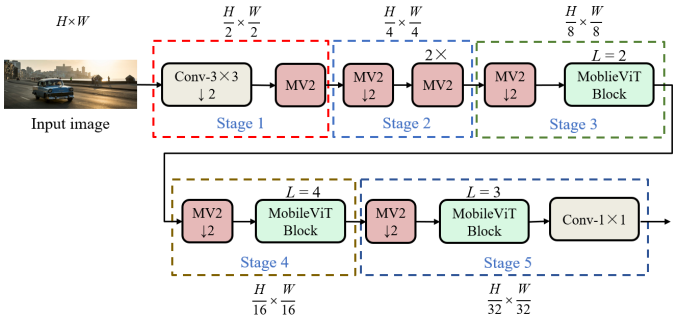


Fig. 4. Student network encoder details. Our encoder is divided into five stages. Each stage outputs a feature map that is concatenated by the decoder.

network predict four scales of inverse depth map based on the input target frame simultaneously. To calculate the distillation loss between the pre-trained MonoViT model and our student depth network, we adopt the logarithm-based paradigm that takes into account the relative differences between the teacher depth and the student depth:

$$L_d = \frac{1}{S}\sum_{i=1}^{S}\frac{1}{N_i}\sum_{j=1}^{N_i}\log\left(|\hat{d}_{ij} - \tilde{d}_{ij}| + 1\right), \tag{9}$$

where N_i represents pixels number in corresponding scale, and $\hat{d}_{ij}$ and $\tilde{d}_{ij}$ represent the teacher inverse depth and the student inverse depth, respectively. By combining distillation loss L_d and baseline loss $L_{baseline}$, we can finally derive the total loss function for our model:

$$L_{total} = L_{baseline} + L_d. \tag{10}$$

C. MViTDepth Architecture

The overview of the proposed MViTDepth network is depicted in Fig. 2. Our training framework consists of three components. The first part involves a teacher network, which is implemented using the pre-trained MonoViT [11] model for knowledge distillation. The weights of the teacher network are fixed during training, and applied to predict a depth map D_{teach}. The second part involves a student network, which is a depth network that predicts a depth map D_{pred} during training by leveraging the knowledge distilled from the pre-trained MonoViT. The third part is a pose network that predicts the relative pose $T_{t\to s}$ between a target frame and a source frame. Subsequently, the predicted depth map D_{pred} and the teacher depth map D_{teach} are used to calculate the distillation loss. Additionally, we combine the predicted depth map D_{pred} and the relative pose transformation $T_{t\to s}$ to generate a synthesized image, which then allows to calculate the reconstruction loss between the target frame I_t and the warped image $I_{s\to t}$.

As shown in Fig. 3, the depth network employs a U-shaped encoder-decoder architecture in a coarse-to-fine manner to produce the multi-scale depth maps from an RGB image. Fig. 4 illustrates how we divide the MobileViT backbone network into five stages. Since feature map has a relatively high resolution in the early stages, we are able to use the convolution operations exclusively to extract features by down-sampling it, in order to reduce computational overhead. As for the last three stages, we also introduce the transformer modules that enables us to perceive long-range global information to some extent while still maintaining a smaller parameter and computational load. MobileNetv2 (MV2) module (see details in [34]) adopts depthwise separable convolution, which incurs lower calculation and parameter costs than the traditional convolution. Meanwhile, MobileViT Block (refer to [21] for details) leverages a strategy that first extracts local features, then perceives long-range dependencies, and eventually fuses local and global features to generate superior feature maps.

For the depth network decoder, different from introducing the additional modules or using a complicated up-sampling approach, we only use the bilinear up-sampling to increase the scale of feature maps in order to alleviate the model parameters and computational burden. Then, we concatenate the up-sampled feature map with the corresponding resolution feature map from the encoder into the convolutional layer, as shown in Fig. 3. The decoder outputs four scales of inverse depth map at full, $\frac{1}{2}$, $\frac{1}{4}$, and $\frac{1}{8}$ resolution, respectively. The number of channels in the output feature maps (i.e. parameters) of the decoder is a critical factor. We made various adjustments to these channels, and the results are listed in Table VI.

Our approach utilizes the pose network for relative pose estimation proposed in [10]. Specifically, a pre-trained ResNet18 is implemented as the pose encoder and input a couple of adjacent frames. Furthermore, a pose decoder consisting of four convolutional layers is adopted to predict the corresponding relative pose with 6-DoF between the consecutive images.

IV. Experimental Results

In this section, we first comprehensively evaluate our proposed MViTDepth architecture on the KITTI dataset [35]. We then test the generalization of our method on the Make3D dataset [36]. Next, we conduct various ablation experiments to verify the validity of each component of the proposed architecture. Finally, we provide models in different sizes to accommodate various edge devices.

A. Datasets and Performance Metrics

The KITTI dataset is commonly used for MDE due to its diverse range of driving situations. Before training, we elimi-

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TABLE I
COMPARISON OF MVITDEPTH WITH SOME RECENT REPRESENTATIVE METHODS ON KITTI BENCHMARK USING THE EIGEN SPLIT. THE BEST SCORES ARE IN **BOLD**, AND THE SECOND-BEST SCORES ARE UNDERLINE. THE RESOLUTION MEANS THE INPUT RGB SIZE, NAMELY, "HEIGHT" TIMES "WIDTH". THE ABBREVIATION "M" REFERS TO: KITTI MONOCULAR VIDEOS, "M+SEM": MONOCULAR VIDEOS + SEMANTIC SEGMENTATION.

Method	Data	Resolution	Depth Error (↓)				Depth Accuracy (↑)			#Params. (↓)
			Abs Rel	Sq Rel	RMSE	RMSE log	δ_1	δ_2	δ_3	
Monodepth2-Res18 [10]	M	192 × 640	0.115	0.903	4.863	0.193	0.877	0.959	0.981	14.3M
Monodepth2-Res50 [10]	M	192 × 640	0.110	0.831	4.642	0.187	0.883	0.962	0.982	32.5M
SGDepth [13]	M+Sem	192 × 640	0.113	0.835	4.693	0.191	0.879	0.961	0.981	16.3M
Johnston et al. [37]	M	192 × 640	0.111	0.941	4.817	0.189	0.885	0.961	0.981	14.3M+
CADepth-Res18 [16]	M	192 × 640	0.110	0.812	4.686	0.187	0.882	0.962	0.983	18.8M
HR-Depth [12]	M	192 × 640	0.109	0.792	4.632	0.185	0.884	0.962	0.983	14.7M
Lite-HR-Depth [12]	M	192 × 640	0.112	0.767	4.557	0.184	0.876	0.963	**0.984**	3.1M
R-MSFM3 [30]	M	192 × 640	0.114	0.815	4.712	0.193	0.876	0.959	0.981	3.5M
R-MSFM6 [30]	M	192 × 640	0.112	0.806	4.704	0.191	0.878	0.960	0.981	3.8M
VADepth [15]	M	192 × 640	**0.104**	0.774	4.552	0.181	**0.892**	**0.965**	0.983	18.8M
MonoFormer [19]	M	192 × 640	**0.104**	0.846	4.580	0.183	0.891	0.962	0.982	23.9M+
Lite-Mono-tiny [20]	M	192 × 640	0.110	0.837	4.710	0.187	0.880	0.960	0.982	2.2M
Lite-Mono-small [20]	M	192 × 640	0.110	0.802	4.671	0.186	0.879	0.961	0.982	2.5M
Lite-Mono [20]	M	192 × 640	0.107	**0.765**	4.561	0.183	0.886	0.963	0.983	3.1M
MViTDepth-tiny (Ours)	M	192 × 640	0.111	0.825	4.731	0.187	0.881	0.961	0.983	**1.8M**
MViTDepth-small (Ours)	M	192 × 640	0.107	0.788	4.579	0.183	0.887	0.963	0.983	2.8M
MViTDepth (Ours)	M	192 × 640	0.106	0.766	**4.507**	**0.180**	0.891	0.964	**0.984**	6.3M
MonoViT (**Teacher**) [11]	M	192 × 640	0.099	0.708	4.372	0.175	0.900	0.967	0.984	27.0M+
Monodepth2-Res18 [10]	M	320 × 1024	0.115	0.882	4.701	0.190	0.879	0.961	0.982	14.3M
R-MSFM3 [30]	M	320 × 1024	0.112	0.773	4.581	0.189	0.879	0.960	0.982	3.5M
R-MSFM6 [30]	M	320 × 1024	0.108	0.748	4.470	0.185	0.889	0.963	0.982	3.8M
Lite-HR-Depth [12]	M	320 × 1024	0.105	**0.704**	**4.374**	0.178	0.891	**0.967**	**0.985**	3.1M
Lite-Mono-tiny [20]	M	320 × 1024	0.104	0.764	4.487	0.180	0.892	0.964	0.983	2.2M
Lite-Mono-small [20]	M	320 × 1024	0.103	0.757	4.449	0.180	0.894	0.964	0.983	2.5M
Lite-Mono [20]	M	320 × 1024	**0.102**	0.746	4.444	0.179	0.896	0.965	0.983	3.1M
MViTDepth-tiny (Ours)	M	320 × 1024	0.109	0.818	4.603	0.184	0.887	0.963	0.983	**1.8M**
MViTDepth-small (Ours)	M	320 × 1024	0.104	0.772	4.460	0.180	0.893	0.965	0.984	2.8M
MViTDepth (Ours)	M	320 × 1024	**0.102**	0.756	4.415	**0.177**	**0.899**	0.966	0.984	6.3M
MonoViT (**Teacher**) [11]	M	320 × 1024	0.096	0.714	4.292	0.172	0.908	0.968	0.984	27.0M+

nate the stationary frames from the raw KITTI. Eventually, we have 39810 monocular triplets for training and 4424 images for validation as in [23]. The performance of our model is tested on the eigen test [4] using 697 images and the corresponding ground truth obtained by LiDAR in order to compare it with the latest SOTA algorithms. The generalization ability of the proposed method is tested on the Make3D dataset, which comprises of 134 outdoor scene test images. Note that the MViTDepth network trained on the KITTI dataset is loaded and directly applied to these test images for inference.

The performance of our proposed model is evaluated by the standard metrics presented in [4], consisting of absolute relative difference (Abs Rel), square related difference (Sq Rel), root mean square error (RMSE), log RMSE, and $\delta_i < 1.25^i$ with $i \in \{1, 2, 3\}$. The scale factor $\hat{s}$ between the predicted depth map and ground truth is defined by

$$\hat{s} = \text{median}(D_{gt})/\text{median}(D_{pred}), \tag{11}$$

where D_{gt} and D_{pred} represent the ground truth and the predicted depth map, respectively. We can thus evaluate the performance by multiplying the predicted depth map D_{pred} with a scaling factor $\hat{s}$ to match the ground truth.

B. Implementation Details

Our model is implemented on a single Nvidia GeForce RTX 3090 platform with Pytorch. Following Monodepth2 [10], we trained our models for 20 epochs using Adam [38] as the optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The learning rate lr is set to $1e^{-4}$ for the first 15 epochs, then it reduces to $1e^{-5}$ for the last 5 epochs. Unlike Monodepth2 that utilizes ResNet [17] backbone, our depth backbone network is based on MobileViT [21]. It should be pointed out that the MobileViT is initialized using the pre-trained ImageNet [39] weights. Our source code and models are available at link.[1]

C. Evaluation on Benchmark Datasets

Table I lists the quantitative results of various prominent self-supervised methods for MDE on KITTI dataset. Compared to other methods using model parameters less than 33M, our proposed MViTDepth exhibits satisfactory results. Specifically, the proposed MViTDepth significantly outperforms Monodepth2-Res18 [10] having more model parameters. It also surpasses Monodepth2-Res50 that has a model size of 32.5M. Moreover, the proposed MViTDepth performs better than the latest MonoFormer [19] in all metrics except for Abs Rel, even though the latter uses a ResNet50 backbone. More importantly, our proposed MViTDepth is more outstanding than the recently proposed lightweight models such as Lite-HR [12], R-MSFM6 [30], and Lite-mono [20] on certain metrics like $\delta < 1.25$. Correspondingly, the quantitative comparison of depth maps among these four lightweight models is shown in Fig. 5.

The performance of our model is also evaluated in terms of model complexity and inference speed. It can be observed from Table II that our MViTDepth has small model parameters and low FLOPs among the considered methods. The results in Table II also show that Monodepth2 based on CNN with

[1][Online]. Available: https://github.com/mengmengbi/MViTDepth

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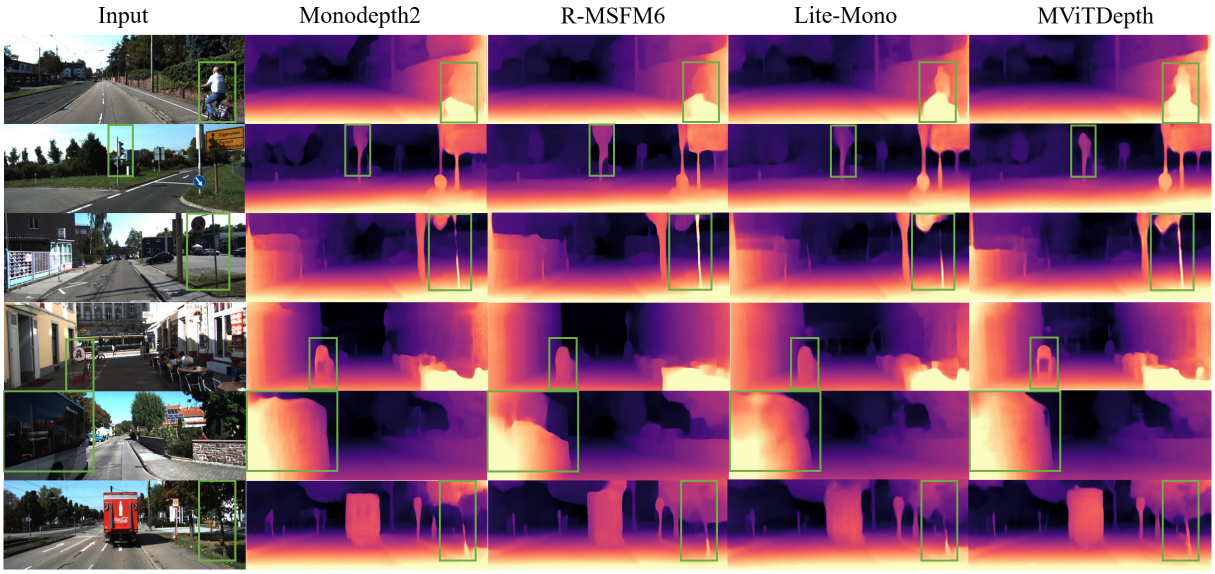


Fig. 5. Qualitative results on the Eigen split of KITTI dataset. Compared to other models [10], [20], [30], our model can preserve more details and sharper edges in depth maps. The differences are highlighted within the green boxes.

TABLE II
QUANTITATIVE COMPARISON OF THE DISTRIBUTIONS OF PARAMETERS, FLOPS, AND INFERENCE SPEED. WE USE THE 192×640 AS IMAGE INPUT SIZE WITH A BATCH SIZE OF 16. TESTING SPEED IS OBTAINED BY AVERAGING OVER 300 TIMES. ALL THE RESULTS ARE TESTED ON A SINGLE RTX 3090.

Method	Encoder			Decoder			Full Model		
	Params.	FLOPs	Speed	Params.	FLOPs	Speed	Params.	FLOPs	Speed
Monodepth2 [10]	11.2M	4.5G	0.8ms	3.1M	3.5G	0.9ms	14.3M	8.0G	**1.7ms**
R-MSFM3 [30]	0.7M	2.4G	0.2ms	2.8M	14.1G	3.4ms	3.5M	16.5G	3.6ms
R-MSFM6 [30]	0.7M	2.4G	0.2ms	3.1M	28.8G	5.6ms	3.8M	31.2G	5.8ms
Lite-Mono-tiny [20]	2.0M	2.4G	1.6ms	0.2M	0.5G	0.2ms	2.2M	2.9G	1.8ms
Lite-Mono-small [20]	2.3M	4.1G	2.0ms	0.2M	0.7G	0.2ms	2.5M	4.8G	2.2ms
Lite-Mono [20]	2.9M	4.4G	2.1ms	0.2M	0.7G	0.2ms	3.1M	5.1G	2.3ms
MViTDepth-tiny (**Ours**)	1.0M	0.7G	1.5ms	0.8M	0.8G	0.3ms	**1.8M**	**1.5G**	1.8ms
MViTDepth-small (**Ours**)	1.9M	1.8G	1.9ms	0.9M	1.0G	0.4ms	2.8M	2.8G	2.3ms
MViTDepth (**Ours**)	5.0M	3.6G	2.2ms	1.3M	1.1G	0.4ms	6.3M	4.7G	2.6ms

TABLE III
QUANTITATIVE RESULTS ON MAKE3D DATASET. ALL OF THESE MODELS ARE TRAINED ON THE KITTI, AND THEN EVALUATED ON THE MAKE3D.

Method	Depth Error (↓)			
	Abs Rel	Sq Rel	RMSE	RMSE log
Monodepth2 [10]	0.322	3.589	7.417	0.163
R-MSFM6 [30]	0.334	3.285	7.212	0.169
Lite-Mono [20]	0.305	**3.060**	6.981	0.158
MViTDepth (Ours)	**0.298**	3.147	**6.955**	**0.152**
MonoViT (**Teacher**) [11]	0.286	2.758	6.623	0.147

relatively high model parameters and FLOPs exhibits a faster inference speed on RTX 3090 compared to those models based on the multi-head attention mechanism. Nevertheless, the inference speed of our proposed MViTDepth is also relatively fast and comparable to the latest lightweight model Lite-Mono.

The Make3D dataset is used to evaluate the generalization capability of our MViTDepth in different outdoor scenes. The proposed MViTDepth is first trained on the KITTI dataset, using the input images with the resolution of 192×640, in accordance with Monodepth2. Then, the generalization performance of our proposed MViTDepth is investigated on the Make3D dataset, without applying any additional fine-tuning processes. The quantitative results on Make3D in Table III show that the proposed MViTDepth achieves better generalization performance than the other three models.

D. Ablation Study

The ablation study is implemented on the KITTI dataset to validate the effectiveness of each component of the proposed MViTDepth. The ablation study of the proposed method on the KITTI dataset is listed in Table IV. We first assess the accuracy and model complexity of the baseline model, namely, Monodepth2 based on ResNet18, as shown in Table IV. As shown in Fig. 4, the backbone network of Monodepth2 is replaced by the MobileViT backbone network to validate its effectiveness. In order to guarantee the fairness, we retain the decoder from the baseline architecture. By utilizing the MobileViT backbone network, the results of the first four rows in Table IV show that both model performance and complexity are improved compared to Monodepth2-Res18. Despite the lower performance of the second row compared to the baseline, it is still acceptable considering the model size. The validity of knowledge distillation is also confirmed by adding it on top of the previous step. By examining the results of the last three rows in Table IV, it is evident that the integration of knowledge distillation can lead to a significant improvement on standard metrics compared to those counterparts without using it. When all its components are integrated, the proposed MViTDepth outperforms the baseline comprehensively in terms of accuracy and model complexity. Fig. 6 illustrates the qualitative results

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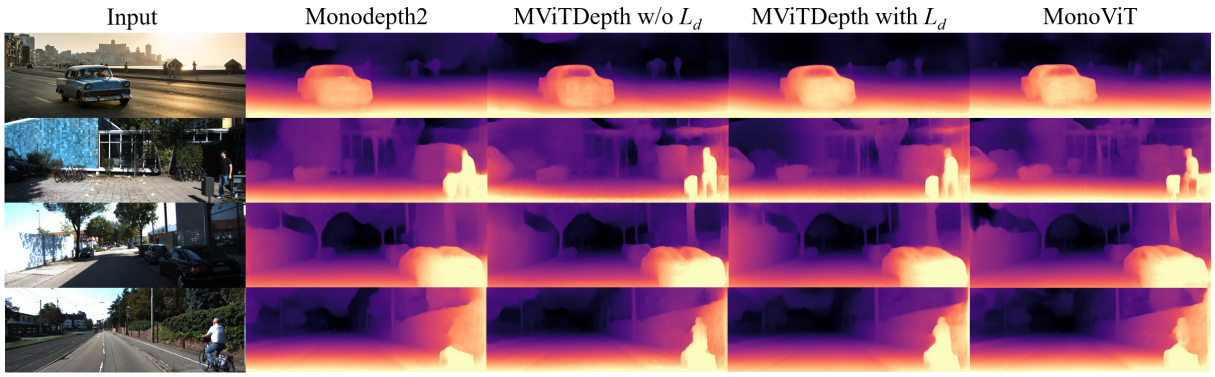


Fig. 6. Qualitative results for ablation study of the backbone and knowledge distillation. Compared to the ResNet-based backbone (column 2), the MobileViT-based backbone (column 3) can extract better features for depth maps and generate good results by suppressing noise and focusing on the key areas. MViTDepth (column 4) utilizing the knowledge from teacher model MonoViT (column 5) can further improve its performance leading to better results.

TABLE IV
ABLATION STUDY OF MVITDEPTH ON THE KITTI DATASET. IN ABLATION STUDY ALL THE DECODERS ARE SAME AS MONODEPTH2 DESCRIBED IN ROW 1. TRAINING AND TESTING ARE BOTH CONDUCTED WITH THE RESOLUTION OF 192×640.

Backbone	$L_{baseline}$	L_d	Depth Error (↓)				Depth Accuracy (↑)			Full Model	
			Abs Rel	Sq Rel	RMSE	RMSE log	δ_1	δ_2	δ_3	Params.	FLOPs
ResNet18 [10]	✓		0.115	0.903	4.863	0.193	0.877	0.959	0.981	14.3M	8.0G
MobileViT-xxsmall [21]	✓		0.119	0.950	4.914	0.196	0.871	0.958	0.981	3.1M	3.2G
MobileViT-xsmall [21]	✓		0.114	0.854	4.716	0.190	0.880	0.961	0.982	4.3M	4.7G
MobileViT-small [21]	✓		0.112	0.833	4.689	0.189	0.883	0.962	0.982	8.1M	6.7G
MobileViT-xxsmall [21]		✓	0.116	0.923	4.895	0.193	0.875	0.959	0.981	3.1M	3.2G
MoblieViT-xsmall [21]		✓	0.107	0.780	4.553	0.182	0.886	0.963	**0.984**	4.3M	4.7G
MobileViT-small [21]		✓	0.108	0.813	4.630	0.184	0.888	0.963	0.983	8.1M	6.7G
MobileViT-xxsmall [21]	✓	✓	0.110	0.829	4.675	0.187	0.884	0.962	0.982	3.1M	3.2G
MobileViT-xsmall [21]	✓	✓	0.106	0.785	4.555	0.182	0.889	0.963	0.983	4.3M	4.7G
MobileViT-small [21]	✓	✓	**0.105**	**0.753**	**4.474**	**0.180**	**0.892**	**0.965**	**0.984**	8.1M	6.7G

TABLE V
ABLATION STUDY OF FROZEN OR UNFROZEN POSE NETWORK ON KITTI DATASET. THE WEIGHTS ARE INITIALIZED WITH THE POSE NETWORK OF MONOVIT, AND ALL THE DECODERS ARE THE SAME AS THOSE IN MONODEPTH2.

Backbone	Resolution	Pose Network	Depth Error (↓)				Depth Accuracy (↑)		
			Abs Rel	Sq Rel	RMSE	RMSE log	δ_1	δ_2	δ_3
MobileViT-xxsmall [21]	192×640	Frozen	0.116	0.891	4.801	0.192	0.877	0.960	0.982
MoblieViT-xsmall [21]	192×640	Frozen	0.112	0.829	4.679	0.188	0.882	0.961	0.982
MobileViT-small [21]	192×640	Frozen	0.110	0.784	4.572	0.185	0.886	0.963	0.983
MobileViT-xxsmall [21]	192×640	Unfrozen	0.110	0.829	4.675	0.187	0.884	0.962	0.983
MoblieViT-xsmall [21]	192×640	Unfrozen	0.106	0.785	4.555	0.182	0.889	0.963	0.983
MobileViT-small [21]	192×640	Unfrozen	**0.105**	**0.753**	**4.474**	**0.180**	**0.892**	**0.965**	**0.984**

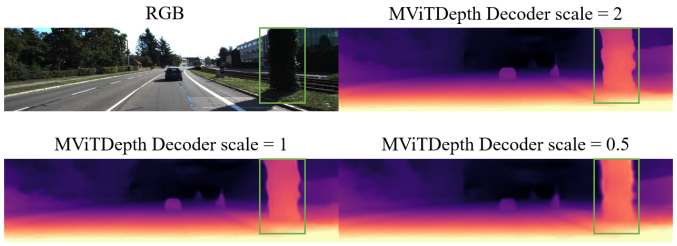


Fig. 7. Qualitative comparison of different number of channels for depth map. These results indicate that the artifacts are reduced in the green boxes when the number of channels is increased in each decoder stage. Note that MobileViT-small is used as the encoder model in this experiment.

of the depth maps for the ablation study. A pose network can be obtained during the pre-training of the teacher network or MonoViT, then its weights can be frozen and used as the pose network for our student network. The performance difference between the frozen pose network and an unfrozen pose network (retrained or fine-tuned) is presented in Table V. It can be observed that the unfrozen pose network yields better results. We also provide various models of different sizes for edge devices to ensure adequate customization for specific requirements. We can adjust the number of feature map channels of decoder output for each layer, allowing for more flexible adjustments to model size. Note that the numbers of feature map channels of the original decoder output in each layer are corresponding to $\{8, 16, 32, 64, 128\}$. As shown in Table VI, with the parameters of the decoder being reduced, its performance also degrades while the corresponding inference time becomes shorter. Fig. 7 shows that the result of depth map becomes more blurred as the number of channels decreases in each layer. Therefore, experimental results demonstrate that our designed MViTDepth can achieve better trade-off between accuracy of depth estimation and model complexity, which is more appropriate to be embedded in edge devices.

V. CONCLUSION

In this letter, we presented a novel lightweight framework of self-supervised MDE by making a trade-off between performance and parameters. The proposed MViTDepth is developed by utilizing MobileViT depth network as its backbone, and transferring knowledge from a pre-trained MonoViT to depth

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TABLE VI
PERFORMANCE COMPARISON OF OUR MODELS IN DIFFERENT SIZES ON KITTI DATASET. THE DECODER CHANNEL SCALE EQUALS 1 CORRESPONDING TO $\{8, 16, 32, 64, 128\}$. TWICE THE NUMBER OF DECODER CHANNEL SCALE CORRESPONDS TO $\{16, 32, 64, 128, 256\}$, AND HALF THE NUMBER CORRESPONDS TO $\{4, 8, 16, 32, 64\}$. TRAINING AND TESTING ARE BOTH CONDUCTED WITH THE RESOLUTION OF 192×640.

Backbone	Decoder Channel Scale	Depth Error (↓)				Depth Accuracy (↑)			Full Model		Speed
		Abs Rel	Sq Rel	RMSE	RMSE log	δ_1	δ_2	δ_3	Params.	FLOPs	
MobileViT-xxsmall [21]	2	0.110	0.829	4.675	0.187	0.884	0.962	0.982	3.1M	3.2G	2.1ms
MobileViT-xsmall [21]	2	0.106	0.785	4.555	0.182	0.889	0.963	0.983	4.3M	4.7G	2.8ms
MobileViT-small [21]	2	**0.105**	**0.753**	**4.474**	**0.180**	**0.892**	**0.965**	**0.984**	8.1M	6.7G	3.1ms
MobileViT-xxsmall [21]	1	0.111	0.825	4.731	0.187	0.881	0.961	0.983	1.8M	1.5G	1.8ms
MobileViT-xsmall [21]	1	0.107	0.788	4.579	0.183	0.887	0.963	0.983	2.8M	2.8G	2.3ms
MobileViT-small [21]	1	0.106	0.766	4.507	**0.180**	0.891	0.964	**0.984**	6.3M	4.7G	2.6ms
MobileViT-xxsmall [21]	0.5	0.114	0.862	4.789	0.189	0.874	0.959	0.983	**1.3M**	**0.9G**	**1.7ms**
MobileViT-xsmall [21]	0.5	0.108	0.755	4.604	0.182	0.883	0.962	**0.984**	2.3M	2.2G	2.2ms
MobileViT-small [21]	0.5	0.106	0.736	4.523	**0.180**	0.888	0.963	**0.984**	5.5M	4.0G	2.5ms

network via offline-distillation. Experimental results show that our approach achieves best balance between accuracy and model complexity, which is particularly useful in resource-constrained application to edge devices.

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